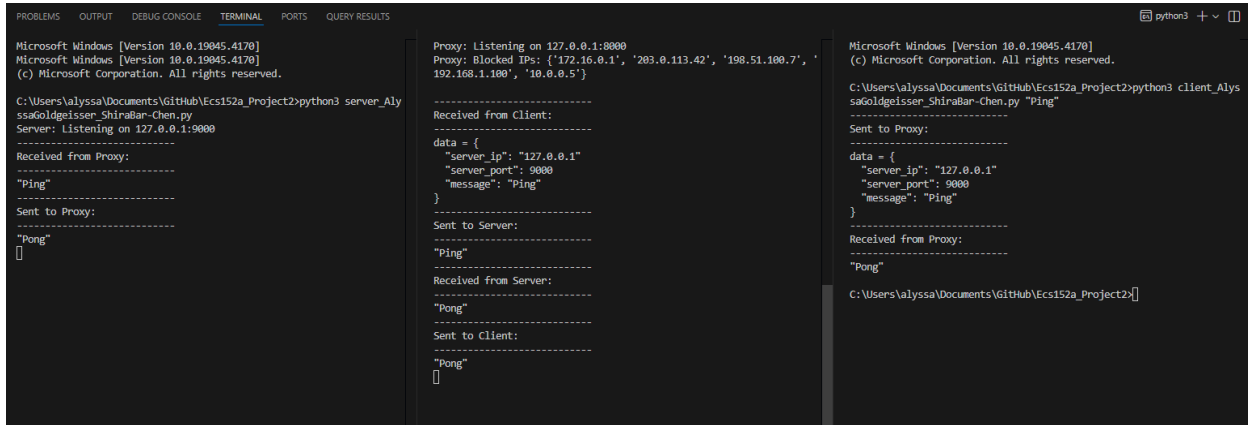


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ECS 152A
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Part 1 Report

1. Print Statements

Ping sent from client:



```
Microsoft Windows [Version 10.0.19045.4170]
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C:\Users\alyssa\Documents\Github\Ecs152a_Project2>python3 server_AlyssaGoldgeisser_ShiraBar-Chen.py
Server: Listening on 127.0.0.1:9000

Received from Proxy:
-----
"Ping"
-----
Sent to Proxy:
-----
"Pong"
[]

Proxy: Listening on 127.0.0.1:8000
Proxy: Blocked IPs: {'172.16.0.1', '203.0.113.42', '198.51.100.7', '192.168.1.100', '10.0.0.5'}

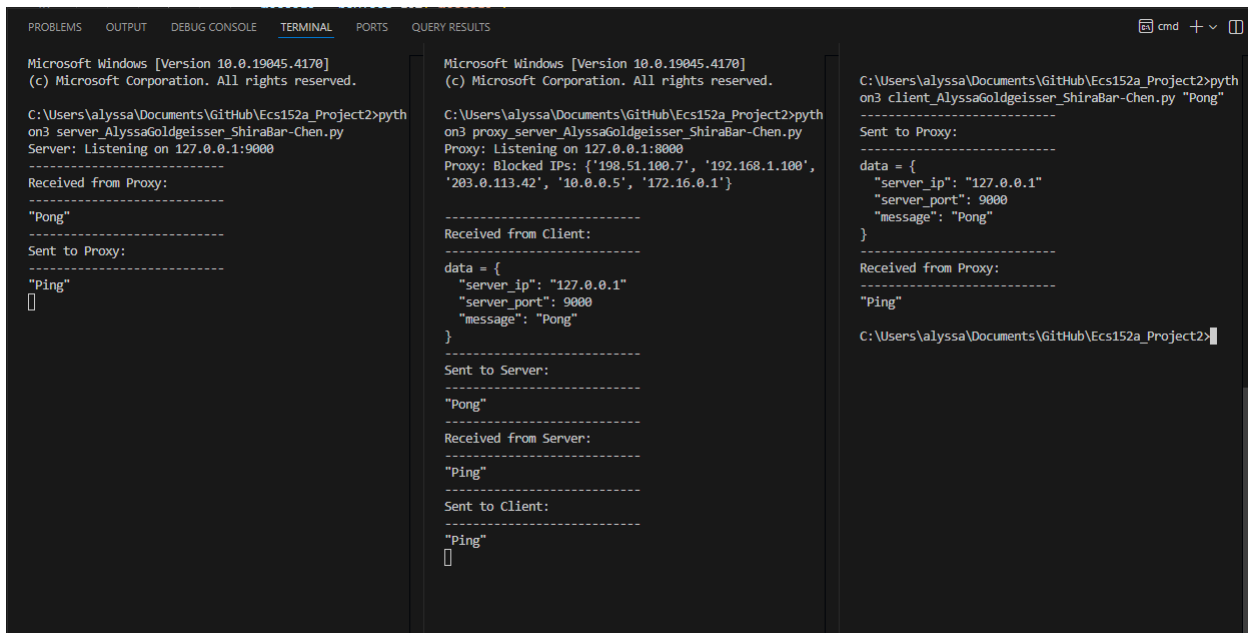
Received from Client:
-----
data = {
  "server_ip": "127.0.0.1"
  "server_port": 9000
  "message": "Ping"
}
-----
Sent to Server:
-----
"Ping"
-----
Received from Server:
-----
"Ping"
-----
Sent to Client:
-----
"Pong"
[]

Microsoft Windows [Version 10.0.19045.4170]
(c) Microsoft Corporation. All rights reserved.

C:\Users\alyssa\Documents\Github\Ecs152a_Project2>python3 client_AlyssaGoldgeisser_ShiraBar-Chen.py "Ping"

Sent to Proxy:
-----
data = {
  "server_ip": "127.0.0.1"
  "server_port": 9000
  "message": "Ping"
}
-----
Received from Proxy:
-----
"Pong"
-----
C:\Users\alyssa\Documents\Github\Ecs152a_Project2>
```

Pong sent from client:



```
Microsoft Windows [Version 10.0.19045.4170]
(c) Microsoft Corporation. All rights reserved.

C:\Users\alyssa\Documents\Github\Ecs152a_Project2>python3 server_AlyssaGoldgeisser_ShiraBar-Chen.py
Server: Listening on 127.0.0.1:9000

Received from Proxy:
-----
"Pong"
-----
Sent to Proxy:
-----
"Ping"
[]

Microsoft Windows [Version 10.0.19045.4170]
(c) Microsoft Corporation. All rights reserved.

C:\Users\alyssa\Documents\Github\Ecs152a_Project2>python3 client_AlyssaGoldgeisser_ShiraBar-Chen.py "Pong"

Sent to Proxy:
-----
data = {
  "server_ip": "127.0.0.1"
  "server_port": 9000
  "message": "Pong"
}
-----
Received from Proxy:
-----
"Ping"
-----
C:\Users\alyssa\Documents\Github\Ecs152a_Project2>
```

Abcd sent from client:

```
Received from Proxy:
-----
"abcd"
-----
Sent to Proxy:
-----
"dcba"
[]

Ping
-----
Sent to Client:
-----
"Ping"
-----
Received from Client:
-----
data = {
  "server_ip": "127.0.0.1"
  "server_port": 9000
  "message": "abcd"
}
-----
Sent to Server:
-----
"abcd"
-----
Received from Server:
-----
"dcba"
-----
Sent to Client:
-----
"dcba"
[]

Ping
-----
C:\Users\alyssa\Documents\GitHub\Ecs152a_Project2>python3 client_AlyssaGoldgeisser_ShiraBar-Chen.py "abcd"
-----
Sent to Proxy:
-----
data = {
  "server_ip": "127.0.0.1"
  "server_port": 9000
  "message": "abcd"
}
-----
Received from Proxy:
-----
"dcba"
-----
C:\Users\alyssa\Documents\GitHub\Ecs152a_Project2>
```

2. How to Run our code

First, to start the server, in a command terminal, run:
`python3 server_AlyssaGoldgeisser_ShiraBar-Chen.py`

Then, create another terminal and run:
`python3 proxy_server_AlyssaGoldgeisser_ShiraBar-Chen.py`

Finally, to start the client open another terminal and run:
`python3 client_AlyssaGoldgeisser_ShiraBar-Chen.py "abcd"` or
`python3 client_AlyssaGoldgeisser_ShiraBar-Chen.py abcd`
Without quotes, both are fine. Where abcd is any four letter string.

3. Description of Expected Result

The system consists of three components, a client, a proxy, and a server, all which communicate with each other over TCP connections.

The client takes a message as an input and sends it to a proxy, along with the target server's IP address and port, packaged as a JSON object. The client never communicates with the server directly, it communicates through the proxy, which forwards the message to the server.

When the proxy receives the request, it first checks if the target's IP address is on the blocklist. If the IP is blocked, the proxy rejects the request and sends a "Blocklist Error" message back to the client, and never actually contacts the server. If the IP is allowed, the proxy continues to forward the message to the server on the client's behalf, along with the proxy's IP address so that the server knows where the request came from. This means that the server never knows the client's IP address, so that the message seems like it came from the proxy.

The server receives the message from the proxy, and processes it according to some rules. In the default case, the message is reversed. If the message is Ping, it gets switched to Pong. If the

message is Pong, it is switched to Ping. Once the message is processed, the server sends the response back to the proxy.

The proxy then forwards this message back to the original client, completing the communication.

Explanation of Initial Attempts:

My main bug had the proxy forwarding only the raw message string to the server, meaning the server had no way of knowing where the request came from. I fixed this by wrapping the proxy's outgoing data in a JSON object, that included the proxy's IP address, and updating the server to parse that JSON and extract it.

I also initially had only the `server_ip` and `message` in the client's JSON payload, but then I added the `server_port` as a field so that the proxy could dynamically connect to the correct port, instead of using a hardcoded default.

When writing the client, server, and proxy, I would often get issues and now know where they came from. I fixed this by adding try except blocks, and on errors, I would sometimes print the error message as `e` so I could look up specifics of what my issue is.